

H-diag (RES-1) off-diagonal floor: the conservative leading stability ratio

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

The constant certificate (hdiag-offdiag-constant-certificate) verified the off-diagonal leading ratio for BCC only ($R_{\text{lead}} = 0.174$). This note extends it to the full enumerated competitor class and exhibits the parallel with RES-5: H-diag and the RES-5 endpoint are governed by the SAME additive-energy quantity K_{floor} . Scope: the $I = 2 \times 10^{-3}$ endpoint; the enumerated crystallographic class; the conservative leading condensate-direction ratio (the operator norm is the standing residual). No tier action.

2. The leading ratio in terms of K_{floor}

The off-diagonal Bogoliubov-Hessian weight is the additive energy $\sum_Q p_2(Q)^2 = E_+$ (proved). The conservative leading condensate-direction ratio is

$$R_{\text{lead}}(I) = \frac{9}{4} \frac{u_{\text{eff}}^2 B_{\text{max}} E_+}{c_{\text{diag}}} \frac{I}{N}, \quad c_{\text{diag}} = \frac{N}{2} r_R,$$

and since $c_{\text{diag}} \propto N$ and $E_+/N^2 = 1 + K_{\text{floor}}$,

$$R_{\text{lead}}(I) = \frac{9}{2} \frac{u_{\text{eff}}^2 B_{\text{max}}}{r_R} (1 + K_{\text{floor}}) I = \text{const} \cdot (1 + K_{\text{floor}}) I, \quad \text{const} = 23.22.$$

R_{lead} is thus controlled by the SAME $K_{\text{floor}} = E_+/N^2 - 1$ as the RES-5 endpoint tail. (BCC: $K_{\text{floor}} = 2.75 \Rightarrow R_{\text{lead}} = 0.174$, reproducing the certificate.)

3. Every enumerated competitor closes

competitor	N	K_{floor}	$R_{\text{lead}}(2 \times 10^{-3})$
$\{100\}, \{200\}$	6	1.50	0.116
$\{111\}$	8	2.38	0.157
$\{110\}$ BCC	12	2.75	0.174
$\{110\} + \{200\}$	18	4.50	0.255
$\{110\} + \{200\} + \{211\}$	42	11.97	0.602

Every enumerated competitor has $R_{\text{lead}} \leq 0.602 < 1$ (the dense 3-shell is the worst). The closure condition is

$$R_{\text{lead}} < 1 \iff K_{\text{floor}} < \frac{1}{\text{const} \cdot I} - 1 = 20.5 \quad (I = 2 \times 10^{-3}),$$

and the enumerated worst $K_{\text{floor}} = 12 < 20.5$. So the diagonal-Gaussian trial is the infimum (the off-diagonal full-covariance dressing does not lower the layer value) over the enumerated class – H-diag is discharged there. Off the endpoint $R_{\text{lead}} \propto I$ shrinks (3-shell $0.602 \rightarrow 0.301$ at 10^{-3}).

4. Parallel with RES-5; class-wide reduction to R-026

H-diag (RES-1) and the RES-5 endpoint are the SAME additive-energy problem viewed twice: both reduce to $K_{\text{floor}} = E_+/N^2 - 1$, with thresholds $K_{\text{floor}} < 20.5$ (RES-1) and < 26.2 (RES-5) – the enumerated competitors ($K_{\text{floor}} \leq 12$) clear both. Class-wide, both reduce to the lattice additive-energy bound R-026 ($E_+ \leq (1 + C_\varepsilon R^\varepsilon)N^2$, T7 modulo textbook NT, decoupling-free) and inherit its residuals. H-LAYER’s two deepest analytic axes therefore collapse onto one machinery.

5. Verdict

H-diag (RES-1) is discharged for the enumerated competitor class (STRONG EVIDENCE): $R_{\text{lead}} < 1$ exactly for every enumerated competitor, so the diagonal-Gaussian trial is the infimum there. **Residuals (unchanged):** (i) the full worst-direction operator norm (vs the conservative leading condensate-direction R_{lead}); (ii) the class-wide constant pin $C_\varepsilon R^\varepsilon + \chi(P) \lesssim T'$ link (the same R-026 items as RES-5). No tier flip: B2/B1 stay T6 CONDITIONAL on {H-LAYER}; the off-diagonal axis of H-LAYER is discharged for the enumerated class, leaving the operator-norm + R-026 residuals.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “ R_{lead} is only the leading condensate-direction ratio, not $\|\cdot\|_{\text{op}}$.” VALID (the standing residual): the conservative $R_{\text{lead}} < 1$ is strong evidence the diagonal trial is the infimum, but the full worst-direction operator norm is not bounded here. The $\times 1.7$ – 8.6 headroom ($R_{\text{lead}} \leq 0.6$) is the cushion against the leading-vs-operator gap; a rigorous operator-norm bound is the named follow-on.
- (β) “The 3-shell 0.602 is close-ish to 1.” VALID-with- mitigation: 0.602 is the conservative leading value at the worst (densest) enumerated competitor at the worst (endpoint) intensity; it stays below unity with a $\times 1.66$ margin, and shrinks off the endpoint. Denser-than-enumerated patterns are the R-026 (lattice) / DR-2 (arbitrary-Q) question, identical to RES-5.
- (γ) “Does this discharge H-LAYER?” No – H-diag/RES-1 is one axis of the H-LAYER hypothesis; this discharges it (strong evidence) for the enumerated class. H-LAYER also rests on the isotropy-infimum theorem (Math427) and the other RES items; this note removes the off-diagonal worry for the actual competitors, not the whole hypothesis. B1/B2 stay T6.

Quantitative sanity check (mandatory): *anchor* $R_{\text{lead}}(\text{BCC}) = 0.174$ reproduces the certificate; *worst* $0.602 < 1$ (3-shell, factor 1.66); *threshold* $K_{\text{floor}} < 20.5$ vs enumerated ≤ 12 ; *scaling* $R_{\text{lead}} \propto I$ ($0.602 \rightarrow 0.301$ at 10^{-3}).

7. Result footer

Result ID:	B2-PROPA-HLAYER / H-diag (RES-1) off-diagonal floor
Precise statement:	The off-diag Bogoliubov weight is E_+ , so the conservative leading condensate-direction ratio is $R_{\text{lead}} = \text{const}(1+K_{\text{floor}})I$, $\text{const} = (9/2)u_{\text{eff}}^2 B_{\text{max}}/rR = 23.22$, $K_{\text{floor}} = E_+/N^2 - 1$. $R_{\text{lead}} < 1$ iff $K_{\text{floor}} < 20.5$ at $I = 2e-3$; every enumerated competitor has $K_{\text{floor}} \leq 12 \rightarrow R_{\text{lead}} \leq 0.602 < 1$ (3-shell worst). Diagonal-Gaussian is the infimum over the enumerated class (H-diag discharged there). Same K_{floor} as RES-5 (threshold 26.2); class-wide reduces to R-026.
Scope:	$I = 2e-3$ endpoint; enumerated crystallographic class; leading condensate-direction ratio (operator norm = residual).
Dependencies:	hdiag-offdiag-additive-energy (E_+ identity); hdiag-offdiag-constant-certificate (B_{max} , c_{diag} , BCC $R_{\text{lead}} = 0.174$); res5-dr2-kappa-bound v1.2 (the K_{floor} / R-026 parallel).
Evidence grade:	ANALYTIC + EXACT (K_{floor} combinatorial) + EXECUTED (res1_hdiag_offdiag_floor.py 4/4).
Reproduction command:	python codes/vacuum/res1_hdiag_offdiag_floor.py
Expected output:	$\text{const} = 23.22$; BCC $R_{\text{lead}} = 0.174$; worst enumerated $0.602 < 1$; threshold $K_{\text{floor}} < 20.5$; 4/4 PASS.
Falsification gate:	An enumerated competitor with $R_{\text{lead}} \geq 1$ ($K_{\text{floor}} > 20.5$); or the full operator norm exceeding 1 where $R_{\text{lead}} < 1$.
Tier before / after:	No tier flip. B2/B1 T6 on {H-LAYER}. H-diag/RES-1 off-diagonal axis DISCHARGED (strong evidence) for the enumerated class; operator norm + R-026 pin = residuals.
No-overclaim statement:	STRONG EVIDENCE, not a theorem: R_{lead} is the conservative leading condensate-direction ratio, not the worst-direction operator norm; the lattice-class constant pin + chi link are R-026 residuals (as RES-5). H-diag is discharged for the enumerated competitors only; H-LAYER (whole) unchanged. B1/B2 T6.
Next required action:	The worst-direction operator-norm bound (complete Hessian) and the R-026 constant pin + chi(P) link (shared with RES-5) are the residuals to make H-diag a theorem over the lattice class.